HOSPITAL MANAGEMENT SYSTEM USING SQL

Tools & Technologies Used :

- MySQL
- SQL Server Management Studio (SSMS)

PROJECT OVERVIEW

- Hospitals generate large amounts of data related to patients, doctors, appointments, treatments, and billing.
- Efficient management of this data is important for improving hospital operations and patient care.
- This project develops a Hospital Management System database using SQL Server to store and manage hospital data in structured tables.
- SQL queries are used to analyze important aspects such as patient demographics, doctor workload, treatment costs, appointment status, and hospital revenue.
- The analysis helps generate useful insights that support better decision-making in hospital management.

OBJECTIVE

- Design a structured relational database for hospital management.
- Store and manage hospital data such as patients, doctors, appointments, treatments, and billing records.
- Establish relationships between tables using primary and foreign keys.
- Use SQL queries to analyze hospital data and answer business-related questions.
- Generate insights related to hospital revenue, doctor workload, treatment costs, and appointment trends.

TABLES IN THE DATABASE

Patients Table:

Column Name	Data Type	Description
patient_id (PK)	VARCHAR(10)	Unique identifier for each patient
first_name	VARCHAR(50)	Patient first name
last_name	VARCHAR(50)	Patient last name
gender	CHAR(1)	Gender of the patient (M/F)
date_of_birth	DATE	Patient's date of birth
contact_number	VARCHAR(15)	Patient contact number
address	VARCHAR(150)	Residential address of the patient
registration_date	DATE	Date when the patient registered
insurance_provider	VARCHAR(50)	Insurance company of the patient

Doctors Table:

Column Name	Data Type	Description
doctor_id (PK)	VARCHAR(10)	Unique identifier for each doctor
first_name	VARCHAR(50)	Doctor's first name
last_name	VARCHAR(50)	Doctor's last name
specialization	VARCHAR(50)	Medical specialization of the doctor
phone_number	VARCHAR(15)	Doctor contact number
years_experience	INT	Total years of medical experience
hospital_branch	VARCHAR(100)	Hospital branch where the doctor works

Appointments Table:

Column Name	Data Type	Description
appointment_id (PK)	VARCHAR(10)	Unique identifier for each appointment
patient_id (FK)	VARCHAR(10)	References patient in the Patients table
doctor_id (FK)	VARCHAR(10)	References doctor in the Doctors table
appointment_date	DATE	Date of the appointment
appointment_time	TIME	Time of the appointment
reason_for_visit	VARCHAR(100)	Reason for the patient visit
status	VARCHAR(20)	Appointment status (Scheduled, Completed, Cancelled)

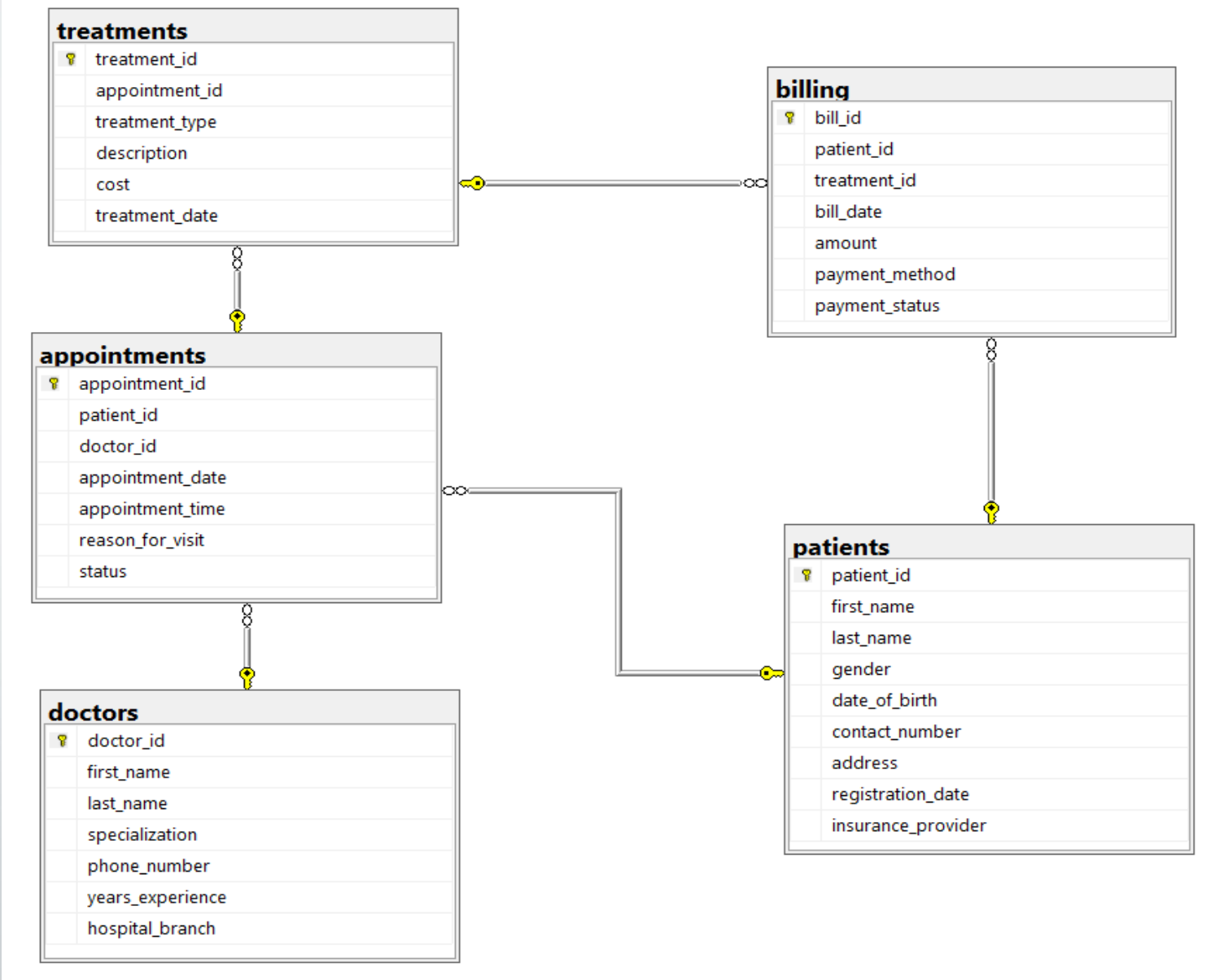
Treatments Table:

Column Name	Data Type	Description
treatment_id (PK)	VARCHAR(10)	Unique identifier for each treatment
appointment_id (FK)	VARCHAR(10)	References appointment in the Appointments table
treatment_type	VARCHAR(50)	Type of treatment provided
description	VARCHAR(150)	Additional details about the treatment
cost	DECIMAL(10,2)	Cost of the treatment
treatment_date	DATE	Date when the treatment was performed

Billing Table:

Column Name	Data Type	Description
bill_id (PK)	VARCHAR(10)	Unique identifier for each bill
patient_id (FK)	VARCHAR(10)	References patient in the Patients table
treatment_id (FK)	VARCHAR(10)	References treatment in the Treatments table
bill_date	DATE	Date when the bill was generated
amount	DECIMAL(10,2)	Total billing amount
payment_method	VARCHAR(30)	Payment method (Cash, Card, Insurance, Online)
payment_status	VARCHAR(20)	Status of payment (Paid, Pending, Failed)

DATABASE DIAGRAM



One-to-Many Relationships :

- **Patients → Appointments (1:M)**

One patient can have multiple appointments, but each appointment belongs to only one patient.

- **Doctors → Appointments (1:M)**

One doctor can handle many appointments, but each appointment is assigned to one doctor.

- **Appointments → Treatments (1:M)**

One appointment may involve multiple treatments, but each treatment record belongs to one appointment.

- **Patients → Billing (1:M)**

One patient can have multiple billing records, but each bill belongs to one patient.

- **Treatments → Billing (1:M)**

One treatment can generate billing information, and billing records are linked to specific treatments.

SQL QUERIES

1. **Total revenue generated**

```sql
SELECT SUM(amount) AS total_revenue
FROM Billing;
```

	total_revenue
1	551249.85

2. Average Treatment Cost

```sql
SELECT CAST(ROUND(AVG(cost),2) AS DECIMAL(10,2))
AS average_treatment_cost
FROM Treatments;
```

	average_treatment_cost
1	2756.25

3. Patient Distribution by Gender

```sql
SELECT gender,COUNT(*) AS total_patients
FROM Patients
GROUP BY gender
ORDER BY total_patients DESC;
```

	gender	total_patients
1	M	31
2	F	19

4. Doctors by Specialization

```sql
SELECT specialization,COUNT(*) AS total_doctors
FROM Doctors
GROUP BY specialization
ORDER BY total_doctors DESC;
```

	specialization	total_doctors
1	Pediatrics	5
2	Dermatology	3
3	Oncology	2

5. Appointment Distribution by Status

```sql
SELECT status,COUNT(*) AS total_appointments
FROM Appointments
GROUP BY status
ORDER BY total_appointments DESC;
```

	status	total_appointments
1	No-show	52
2	Scheduled	51
3	Cancelled	51
4	Completed	46

6. Patient appointments with doctor details

```sql
SELECT p.first_name,
       p.last_name,
       a.appointment_date,
       a.status
FROM Patients p
JOIN Appointments a ON
p.patient_id = a.patient_id
WHERE a.status = 'Scheduled';
```

	first_name	last_name	appointment_date	status
1	Alex	Smith	2023-08-09	Scheduled
2	Linda	Miller	2023-06-19	Scheduled
3	David	Williams	2023-04-09	Scheduled
4	Jane	Wilson	2023-03-05	Scheduled
5	Laura	Jones	2023-08-16	Scheduled
6	Michael	Taylor	2023-06-30	Scheduled
7	Robert	Williams	2023-07-11	Scheduled
8	David	Wilson	2023-11-14	Scheduled
9	Emily	Miller	2023-11-06	Scheduled
10	Michael	Wilson	2023-04-18	Scheduled
11	Emily	Moore	2023-03-28	Scheduled
12	Robert	Williams	2023-02-23	Scheduled
13	Linda	Johnson	2023-04-17	Scheduled
14	Michael	Wilson	2023-03-21	Scheduled
15	Michael	Taylor	2023-09-28	Scheduled
16	Michael	Taylor	2023-12-16	Scheduled
17	Emily	Smith	2023-10-06	Scheduled
18	David	Moore	2023-01-02	Scheduled
19	Laura	Wilson	2023-06-29	Scheduled
20	Linda	Brown	2023-08-10	Scheduled

7. Top 5 Doctors with most appointments

```sql
SELECT TOP 5 d.first_name, d.last_name,COUNT(a.appointment_id) AS total_appointments
FROM Doctors d
JOIN Appointments a
ON d.doctor_id = a.doctor_id
GROUP BY d.first_name, d.last_name
ORDER BY total_appointments DESC;
```

	first_name	last_name	total_appointments
1	Sarah	Taylor	29
2	David	Taylor	25
3	Alex	Davis	24
4	Jane	Smith	22
5	Jane	Davis	21

8. Treatments received by patients

```sql
SELECT p.first_name,
       p.last_name,
       t.treatment_type,
       t.cost
FROM Patients p
JOIN Appointments a
ON p.patient_id = a.patient_id
JOIN Treatments t
ON a.appointment_id = t.appointment_id;
```

	first_name	last_name	treatment_type	cost
1	Alex	Smith	Chemotherapy	3941.97
2	Alex	Moore	MRI	4158.44
3	Emily	Miller	MRI	3731.55
4	Robert	Wilson	MRI	4799.86
5	Emily	Williams	ECG	582.05
6	Linda	Miller	Chemotherapy	1381.00
7	David	Williams	Chemotherapy	534.03
8	Michael	Taylor	Physiotherapy	3413.64
9	Jane	Wilson	Physiotherapy	4541.14
10	David	Wilson	Physiotherapy	1595.67
11	John	Brown	MRI	4671.66
12	David	Smith	Chemotherapy	771.20
13	Laura	Jones	MRI	4704.96
14	Laura	Davis	ECG	2082.30
15	John	Taylor	Physiotherapy	956.39
16	Michael	Taylor	MRI	2686.42
17	Robert	Williams	MRI	1655.49
18	John	Brown	ECG	1781.93
19	David	Smith	X-Ray	1882.80
20	Alex	Taylor	Chemotherapy	4113.62
21	Alex	Moore	X-Ray	2926.23
22	David	Wilson	Physiotherapy	1900.88
23	Jane	Moore	MRI	3246.50
24	David	Moore	ECG	3722.68
25	Emily	Moore	ECG	1726.81

9. Patient billing information

```sql
SELECT p.first_name,
       p.last_name,
       b.amount,
       b.payment_status
FROM Billing b
JOIN Patients p
ON b.patient_id = p.patient_id;
```

	first_name	last_name	amount	payment_status
1	David	Williams	534.03	Failed
2	David	Williams	3249.41	Failed
3	David	Williams	2960.14	Paid
4	David	Williams	975.49	Pending
5	Emily	Smith	616.15	Paid
6	Emily	Smith	1736.63	Failed
7	Emily	Smith	3615.96	Pending
8	Laura	Jones	3231.92	Pending
9	Laura	Jones	4704.96	Paid
10	Michael	Johnson	4550.10	Failed
11	Michael	Johnson	812.41	Pending
12	David	Wilson	2349.63	Pending
13	David	Wilson	3349.18	Paid
14	David	Wilson	1595.67	Paid
15	David	Wilson	1900.88	Failed
16	David	Wilson	1048.49	Pending
17	David	Wilson	1108.25	Failed
18	David	Wilson	3202.67	Failed
19	David	Wilson	4055.14	Failed
20	Alex	Johnson	3228.14	Paid

10. Treatments performed by specialization

```sql
SELECT d.specialization,COUNT(t.treatment_id) AS total_treatments
FROM Doctors d
JOIN Appointments a ON d.doctor_id = a.doctor_id
JOIN Treatments t ON a.appointment_id = t.appointment_id
GROUP BY d.specialization
ORDER BY total_treatments DESC;
```

	specialization	total_treatments
1	Pediatrics	98
2	Dermatology	70
3	Oncology	32

11. Rank doctors by appointment volume

```sql
SELECT d.first_name,d.last_name,
COUNT(a.appointment_id) AS total_appointments,
RANK() OVER (ORDER BY COUNT(a.appointment_id) DESC)
AS doctor_rank
FROM Doctors d
JOIN Appointments a
ON d.doctor_id = a.doctor_id
GROUP BY d.first_name,d.last_name;
```

	first_name	last_name	total_appointments	doctor_rank
1	Sarah	Taylor	29	1
2	David	Taylor	25	2
3	Alex	Davis	24	3
4	Jane	Smith	22	4
5	Jane	Davis	21	5
6	Linda	Wilson	19	6
7	Sarah	Smith	17	7
8	Linda	Brown	16	8
9	David	Jones	14	9
10	Robert	Davis	13	10

12. Top 5 Patients with total spending

```sql
SELECT TOP 5 p.first_name,p.last_name,
(
    SELECT SUM(b.amount)
    FROM Billing b
    WHERE b.patient_id = p.patient_id
) AS total_spent
FROM Patients p
ORDER BY total_spent DESC;
```

	first_name	last_name	total_spent
1	Laura	Davis	30053.08
2	David	Moore	23554.06
3	Michael	Taylor	22967.94
4	Michael	Wilson	21583.56
5	Robert	Wilson	19513.17

13. Running total of hospital revenue

```sql
SELECT bill_date,
SUM(amount) AS daily_revenue,
SUM(SUM(amount)) OVER (ORDER BY bill_date)
AS running_total
FROM Billing
GROUP BY bill_date;
```

	bill_date	daily_revenue	running_total
1	2023-01-01	5561.98	5561.98
2	2023-01-02	4201.76	9763.74
3	2023-01-03	2736.34	12500.08
4	2023-01-05	894.39	13394.47
5	2023-01-06	3729.19	17123.66
6	2023-01-07	3228.14	20351.80
7	2023-01-08	4833.17	25184.97
8	2023-01-11	2593.43	27778.40
9	2023-01-12	4055.14	31833.54
10	2023-01-13	1595.67	33429.21
11	2023-01-15	3489.34	36918.55
12	2023-01-16	3249.41	40167.96
13	2023-01-17	4652.41	44820.37
14	2023-01-20	3615.96	48436.33
15	2023-01-24	3428.95	51865.28
16	2023-01-26	2960.14	54825.42
17	2023-01-28	2512.41	57337.83
18	2023-01-30	1363.40	58701.23
19	2023-02-03	2761.55	61462.78
20	2023-02-04	4550.10	66012.88

14. Appointment Status Category

```sql
SELECT appointment_id,appointment_date,
CASE
    WHEN status = 'Completed' THEN 'Visit Finished'
    WHEN status = 'Scheduled' THEN 'Upcoming Appointment'
    WHEN status = 'Cancelled' THEN 'Missed Appointment'
    ELSE 'Other'
END AS appointment_category
FROM Appointments;
```

	appointment_id	appointment_date	appointment_category
1	A001	2023-08-09	Upcoming Appointment
2	A002	2023-06-09	Other
3	A003	2023-06-28	Missed Appointment
4	A004	2023-09-01	Missed Appointment
5	A005	2023-07-06	Other
6	A006	2023-06-19	Upcoming Appointment
7	A007	2023-04-09	Upcoming Appointment
8	A008	2023-05-24	Missed Appointment
9	A009	2023-03-05	Upcoming Appointment
10	A010	2023-01-13	Visit Finished
11	A011	2023-11-12	Other
12	A012	2023-05-07	Visit Finished
13	A013	2023-08-16	Upcoming Appointment
14	A014	2023-05-25	Missed Appointment
15	A015	2023-01-15	Other
16	A016	2023-06-30	Upcoming Appointment
17	A017	2023-07-11	Upcoming Appointment
18	A018	2023-11-14	Missed Appointment
19	A019	2023-02-06	Missed Appointment
20	A020	2023-12-05	Visit Finished

15. Treatments with Cost Higher Than Average

```sql
SELECT treatment_type,COUNT(*)
AS high_cost_cases
FROM Treatments
WHERE cost > (
    SELECT AVG(cost)
    FROM Treatments
)
GROUP BY treatment_type
ORDER BY high_cost_cases DESC;
```

	treatment_type	high_cost_cases
1	MRI	23
2	Chemotherapy	22
3	X-Ray	20
4	ECG	19
5	Physiotherapy	19

KEY INSIGHTS

- The hospital generated **₹551,249.85 in total revenue.**
- The **average treatment cost is ₹2,756.25.**
- **Male patients (31)** are higher than **female patients (19)** in the dataset.
- **Pediatrics** has the highest number of doctors (5) compared to other specializations.
- **No-show appointments (52)** are the most frequent, indicating missed scheduling opportunities.
- **Scheduled (51)** and **cancelled (51)** appointments show similar appointment management trends.
- **Pediatrics** performs the highest number of treatments, suggesting higher demand in this department.
- Some doctors manage significantly more appointments, showing uneven workload distribution.
- **MRI and Chemotherapy** treatments frequently appear among high-cost procedures, contributing significantly to revenue.

CONCLUSION

This project demonstrates how SQL can be used to analyze hospital operational data to uncover valuable insights related to patient demographics, appointment management, treatment costs, and revenue generation. By analyzing multiple tables and relationships within the database, the study identifies patterns such as high no-show appointment rates, specialization demand, and high-value treatments. These insights can help hospital management improve resource allocation, appointment scheduling efficiency, and financial planning, ultimately supporting better healthcare service delivery and operational performance.

THANK YOU